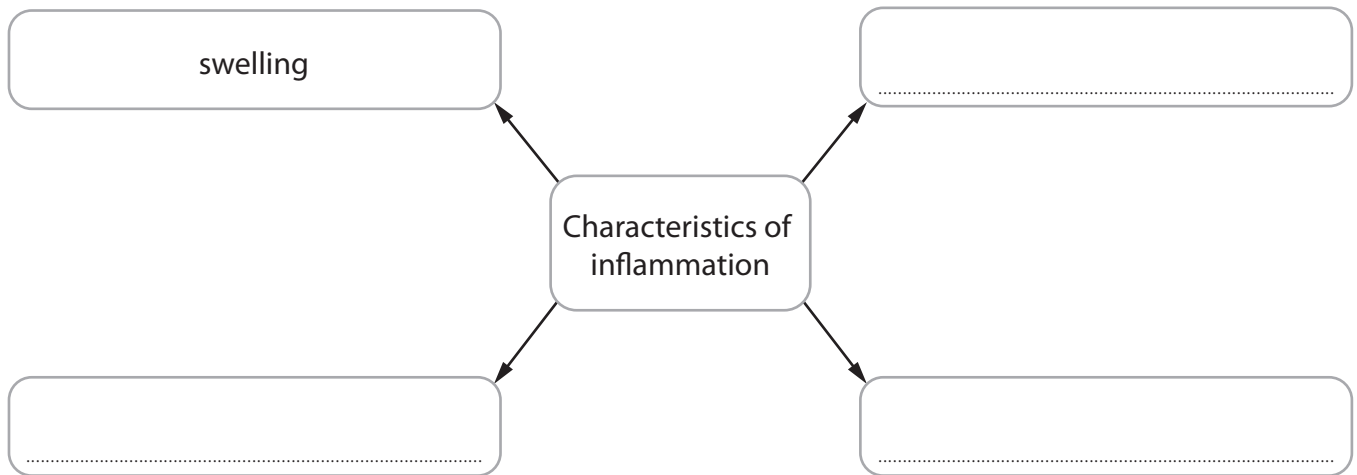


1 Inflammation is involved in the non-specific response of the body to infection and injury.

(a) (i) Complete the diagram to show the four characteristics of inflammation.

(2)



(ii) Inflammation occurs following a cut to the skin.

Describe the role of **two** of these characteristics of inflammation in response to a cut to the skin.

(2)

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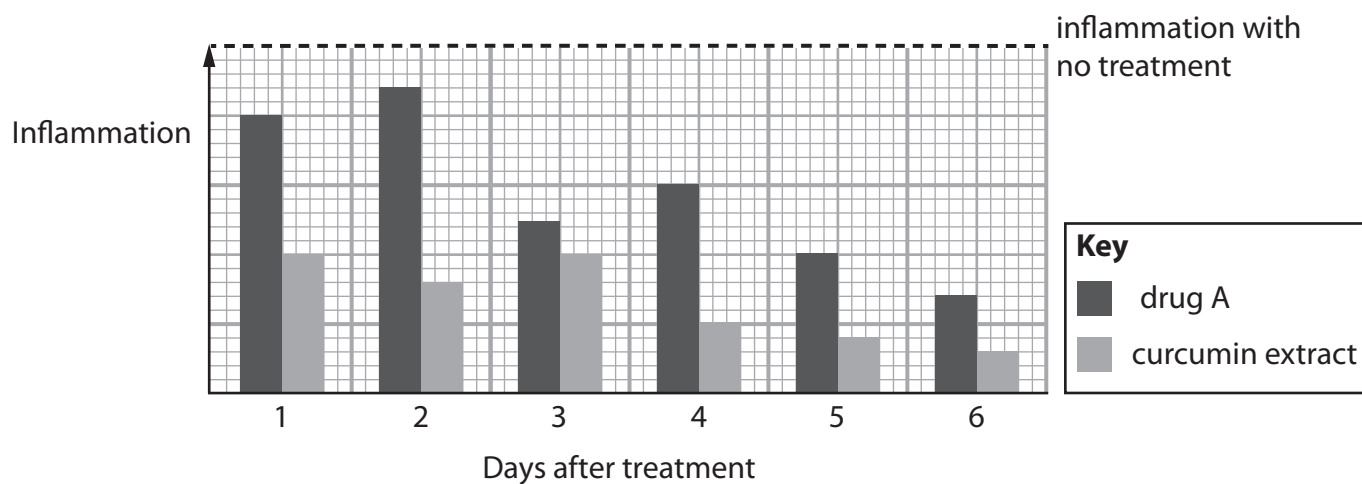
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(b) Anti-inflammatory drugs reduce inflammation.

Turmeric is a spice that is added to food.

Curcumin is a chemical in turmeric that has been shown to reduce inflammation.

- (i) The graph shows the effect of treating inflammation with curcumin extract and an anti-inflammatory drug, drug A.



Compare and contrast the effect of treating inflammation with curcumin extract and with drug A.

(3)

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(ii) Many treatments use 1 g curcumin extract.

A sample of turmeric contains 3 % curcumin.

Which is the mass of turmeric for one treatment?

(1)

- ☐ **A** 0.03 g
- ☐ **B** 3.33 g
- ☐ **C** 33.3 g
- ☐ **D** 33.4 g

(Total for Question 1 = 8 marks)